

Computational Validation of Adaptive Interface Control in TMI Quantum Systems

Extended v5 report on Tests 1–28

Salkutsan Aleksey

May 7, 2026

Abstract

This report extends the TMI computational validation line to Tests 1–28. Version v5 adds Test 28: out-of-distribution Qiskit-Aer generalization on unseen circuit families that were not used in calibration. The main new result is:

$$\text{Calibrate}(I_{sim}) \wedge \text{Correct}(I_{circuit}) \Rightarrow \text{Generalize}(C_{new}).$$

This is simulator validation, not laboratory proof of a physical theory.

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1 Report status

TMI Quantum Computational Validation v0.5

2 Summary of Tests 1-28

Test	Name	Criterion / meaning	Result
1	Ramsey toy model	$T2_TMI > T2_standard$	Delta T2 = 58.1 us
2	Noisy Ramsey learning	$T2_TMI > T2_standard$	Delta T2 = 39.3 us
3	Small quantum circuit	$P_TMI > P_standard$	Delta P = 0.115
4	Stabilized interface	Stable adaptive control	Delta P = 0.026
5	Surrogate Monte Carlo	$E[P_TMI] > E[P_standard]$	Delta = 0.0510, Pr = 0.864
6	Ablation study	Structured adaptive control	full TMI Delta = 0.0680
7	Density-matrix Monte Carlo	Full matrix validation	Delta = 0.0930, Pr = 0.933
8	Cross-model validation	Multiple circuit families	Delta = 0.0660, Pr = 0.846
9	Failure-boundary map	Weak/failure regions	failure = 0, weak = 6
10	Adversarial failure	Forced failure modes	failure cells = 58
11	Safety recovery	Mitigation after failure	recovered = 31/58
12	Reproducibility harness	One-command validation	passed = 11/11
13	Independent simulator	Independent simulator validation	Delta = 0.1121, Pr = 1.000
14	Noise-model stress	Noise class robustness	passed = 3/4, Delta = 0.0068
15	Amplitude damping recovery	Relaxation-aware layer	Delta_relax = 0.1532
16	Noise classifier	Classify then control	accuracy = 0.952, Delta = 0.0554
17	Online noise drift	Dynamic reclassification	Delta = 0.0255, acc = 0.816
18	Delayed feedback	Latency hurts online control	online = 0.883, delayed d2 = 0.861
19	Latency policy optimization	Predictive latency layer	opt = 0.875 > d2 = 0.861
20	Cost-overhead tradeoff	Utility and efficiency	U_highperf = 0.0705

Test	Name	Criterion / meaning	Result
21	Cost-interface boundary	B_cost	lambda* = 0.0636
22	Multi-objective control	Boundary manifold	standard share = 0.608, cost-aware-low = 0.373
23	Standard simulator protocol	Backend-independent Qiskit-style protocol	Delta = 0.0414, Pr = 0.750
24	Raw Qiskit-Aer execution	External simulator without calibration	failed, Delta = -0.00097, cells = 0/16
25	Qiskit noise calibration	Simulator-interface calibration	54 candidate calibrations; Loss_sim = 0.00106
26	Calibrated Qiskit-Aer validation	Held-out calibrated validation	passed, Delta = 0.0725, cells = 12/16
27	Circuit-inverse audit	Correct inverse and re-run Qiskit-Aer	passed, Delta = 0.1144, cells = 16/16
28	OOD Qiskit-Aer generalization	Unseen circuit families	passed, Delta = 0.1252 to 0.1414, cells = 24/24

3 External Qiskit-Aer branch: failure, calibration, correction, generalization

The external branch now has the following structure:

Test 24 : failed $\rightarrow$ *Test 25 : calibrated* $\rightarrow$ *Test 26 : passed* $\rightarrow$ *Test 27 : corrected* $\rightarrow$ *Test 28 : OOD passed*

3.1 Test 24: raw Qiskit-Aer failure

$$P_{base} = 0.7535, \quad P_{standard} = 0.7532, \quad P_{TMI} = 0.7523.$$

$$\Delta(P_{TMI} - P_{standard}) = -0.00097, \quad cells = 0/16.$$

3.2 Test 25: simulator-interface calibration

Test 25 found 54 candidate calibrations and introduced the diagnostic concept:

$$I_{sim} : TMI \text{ model} \rightarrow Qiskit \text{ NoiseModel}, \quad Loss_{sim}(I_{sim}) > 0.$$

In the default-like calibrated regime:

$$\Delta(P_{standard} - P_{base}) = 0.0191, \quad \Delta(P_{TMI} - P_{standard}) = 0.0299.$$

3.3 Test 26: calibrated Qiskit-Aer validation

$$P_{base} = 0.4793, \quad P_{standard} = 0.5203, \quad P_{TMI} = 0.5928.$$
$$\Delta(P_{TMI} - P_{standard}) = 0.0725, \quad cells = 12/16.$$

3.4 Test 27: circuit-inverse audit

Original circuits failed ideal inverse correctness; corrected circuits passed. The key fix was:

$$S \rightarrow Sdg, \quad S^{-1} = S^\dagger.$$

After correction:

$$P_{base} = 0.5791, \quad P_{standard} = 0.6432, \quad P_{TMI} = 0.7576.$$
$$\Delta(P_{TMI} - P_{standard}) = 0.1144, \quad cells = 16/16.$$

3.5 Test 28: out-of-distribution Qiskit-Aer generalization

Test 28 used unseen circuit families, not used in calibration or correction. The conservative uploaded profile gives:

$$P_{base} = 0.2914, \quad P_{standard} = 0.3424, \quad P_{TMI} = 0.4616.$$
$$\Delta(P_{standard} - P_{base}) = 0.0511, \quad \Delta(P_{TMI} - P_{standard}) = 0.1191.$$
$$Pr(P_{TMI} > P_{standard}) = 1.0, \quad cells = 24/24.$$

Across reported OOD profiles, all passed:

best_by_channel, conservative, stress_high_signal.

4 Visual summary

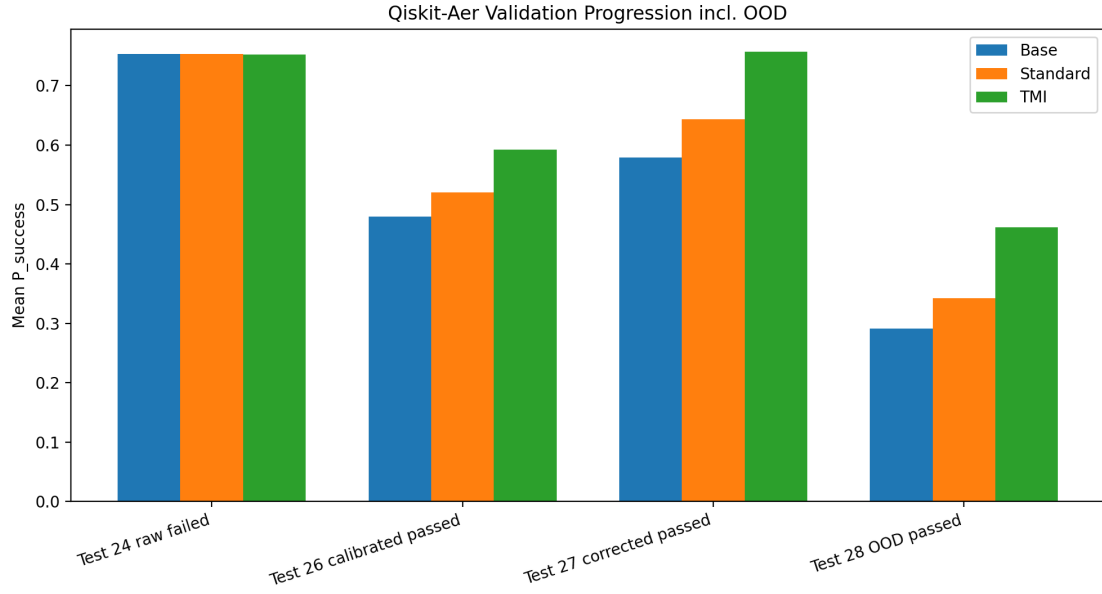


Figure 1: External Qiskit-Aer progression including OOD generalization.

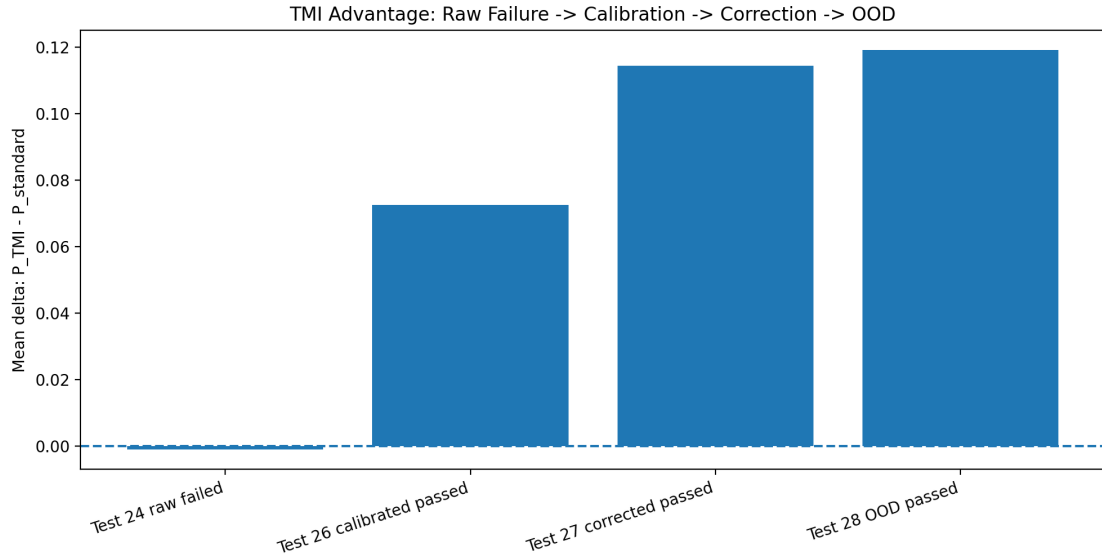


Figure 2: $P_{\text{TMI}} - P_{\text{standard}}$: raw failure becomes positive after calibration, correction and OOD validation.

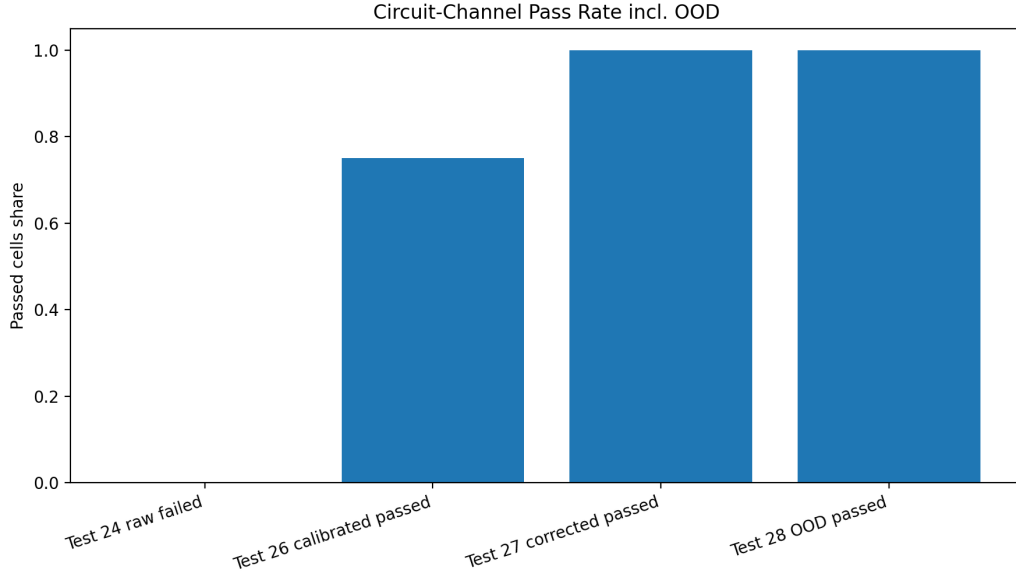


Figure 3: Cell pass rate: 0/16, 12/16, 16/16, 24/24.

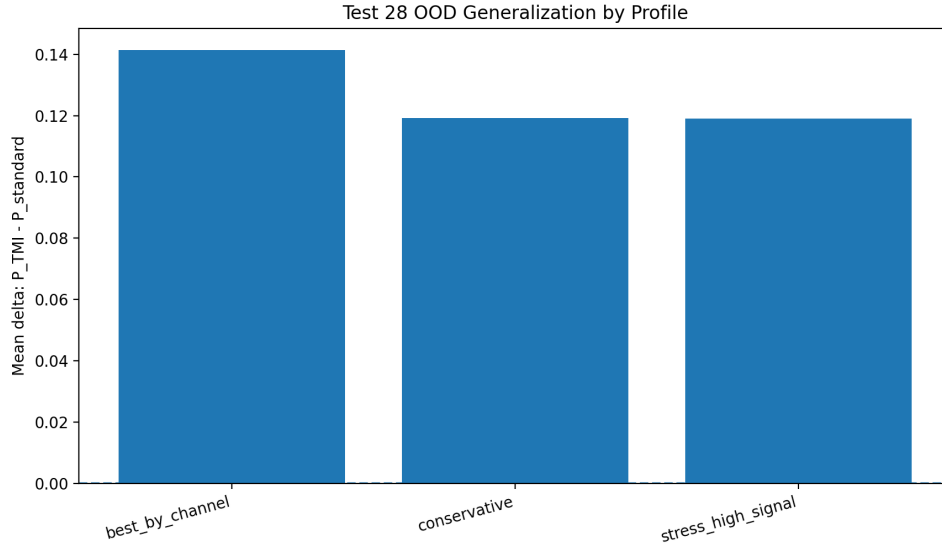


Figure 4: Test 28 OOD deltas by profile.

5 v5 formulas

$$\text{Calibrate}(I_{sim}) \wedge \text{Correct}(I_{circuit}) \Rightarrow \text{Generalize}(C_{new})$$

$$C_{new} \notin C_{calibration} \Rightarrow P_{TMI} > P_{standard} > P_{base}$$

6 Conclusion

Version v5 adds out-of-distribution Qiskit-Aer generalization. The result is stronger than v4 because it shows that the calibrated/corrected simulator-

interface mechanism is not limited to the circuit families used in calibration.

A Package contents

The package contains Russian and English PDF/LaTeX sources, JSON/CSV summaries, external Qiskit-Aer branch figures, Tests 12-28 packages, and Tests 24-28 status files.